EXPERIMENT 6

Application of K-Means and Hierarchical Clustering on Customer Segmentation Dataset

1. Dataset Source

The dataset used for this experiment is:

Customer Segmentation Dataset

Source Platform:
Kaggle

Dataset Link:
https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python

2. Dataset Description

The dataset contains customer information used for market segmentation analysis.

Dataset Size:

- Number of Instances: 200
- Number of Features: 5

Features:

Feature	Description
CustomerID	Unique ID of customer
Gender	Male/Female
Age	Age of customer
Annual Income (k$)	Annual income in thousand dollars

Spending Score (1-100)	Score assigned based on spending behavior

Characteristics:

- No target variable (unsupervised learning)
- Mixed data types (categorical and numerical)
- Suitable for clustering tasks

3. Mathematical Formulation of the Algorithm

K-Means Clustering

K-Means Clustering

K-Means aims to partition data into K clusters by minimizing intra-cluster variance.

Objective Function:

$J=\sum i=1K\sum x\in Ci||x-\mu i||2$J=i=1∑Kx∈Ci∑||x−μi||2

Where:

- CiCi = cluster i
- μiμi = centroid of cluster i
- xx = data point

Steps:

1. Initialize centroids
2. Assign points to nearest centroid
3. Update centroids
4. Repeat until convergence

Hierarchical Clustering

Hierarchical Clustering

In agglomerative clustering:

- Each data point starts as its own cluster

- Clusters are merged step-by-step

Distance between clusters (Ward's Method):

$D(A,B)$=∑$(xi-yi)$2D(A,B)=∑(xi−yi)2

Dendrogram is used to visualize cluster merging.

4. Algorithm Limitations

K-Means Limitations:

- Requires predefined number of clusters (K)
- Sensitive to initial centroid selection
- Not suitable for non-spherical clusters
- Sensitive to outliers

Hierarchical Clustering Limitations:

- Computationally expensive for large datasets
- Difficult to determine optimal clusters
- Once merged, clusters cannot be split
- Sensitive to noise and outliers

5. Methodology / Workflow

Steps Performed:

1. Data Collection
   Dataset downloaded from Kaggle
2. Data Preprocessing
   - Selected relevant features
   - Applied feature scaling
3. K-Means Clustering
   - Used Elbow Method to find optimal K
   - Applied K-Means algorithm
4. Hierarchical Clustering
   - Generated dendrogram
   - Applied Agglomerative Clustering
5. Visualization

- Cluster plots
- Dendrogram

Workflow Diagram

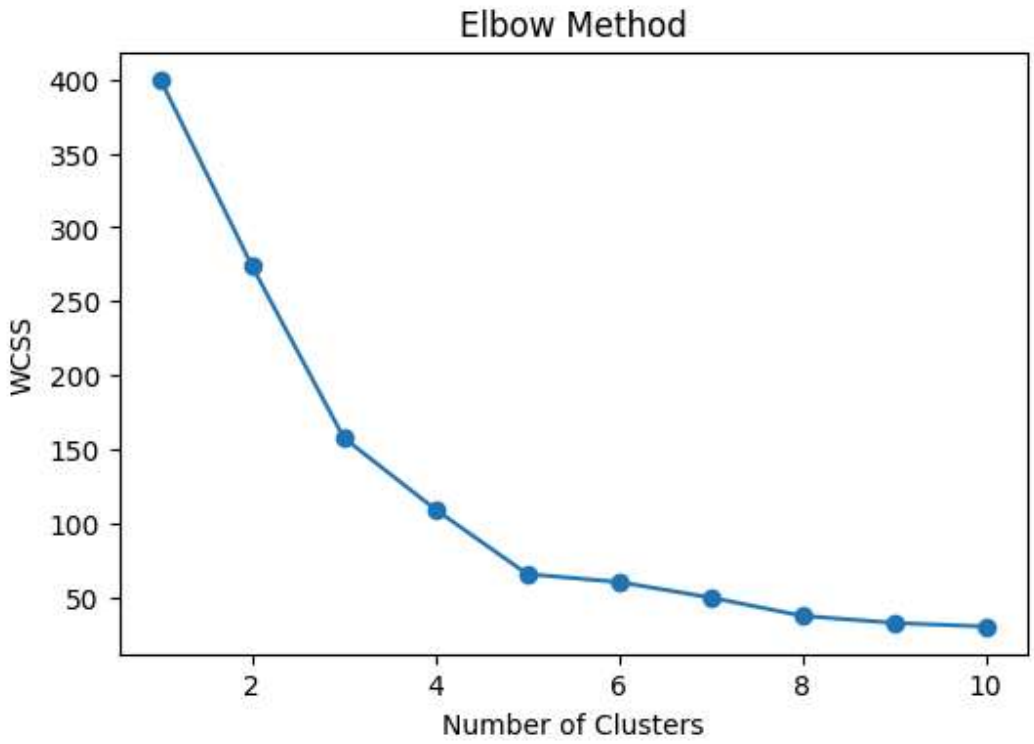
Dataset
↓
Data Preprocessing
↓
Feature Selection
↓
Scaling
↓
K-Means (Elbow Method → Clustering)
↓
Hierarchical Clustering (Dendrogram)
↓
Visualization & Analysis

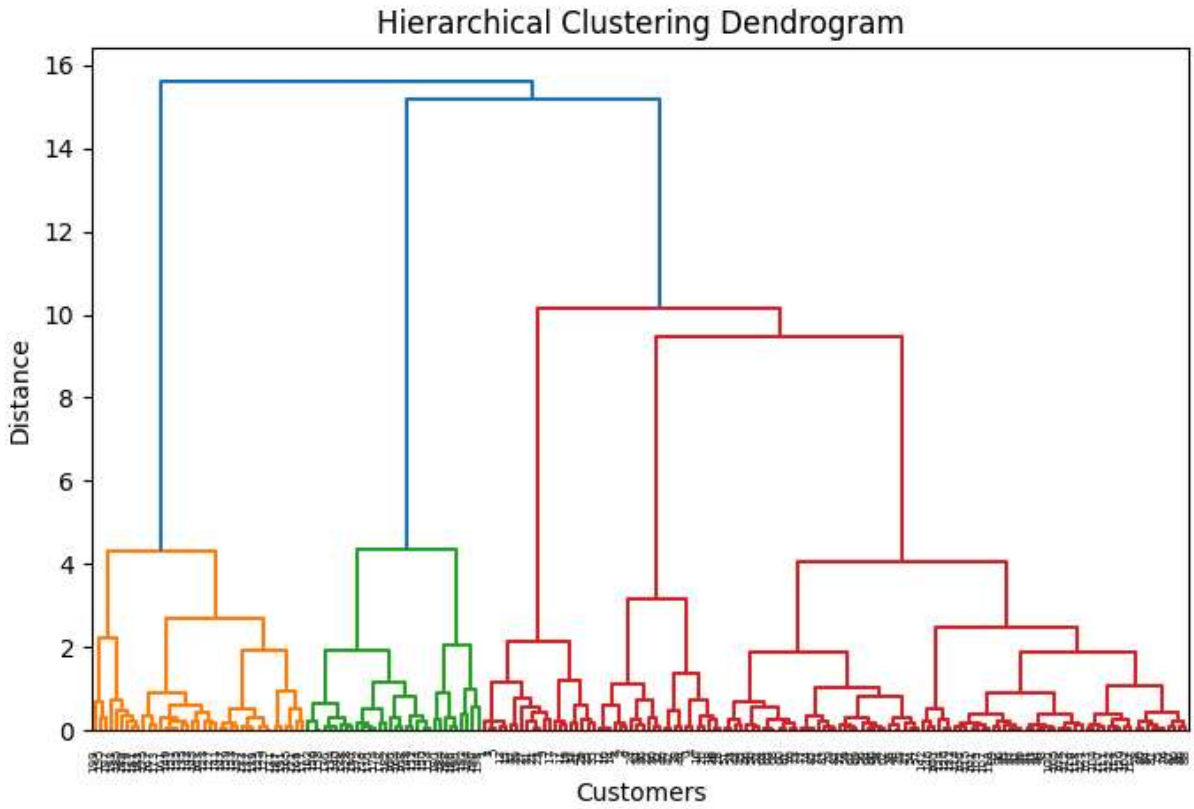
Implementation Code

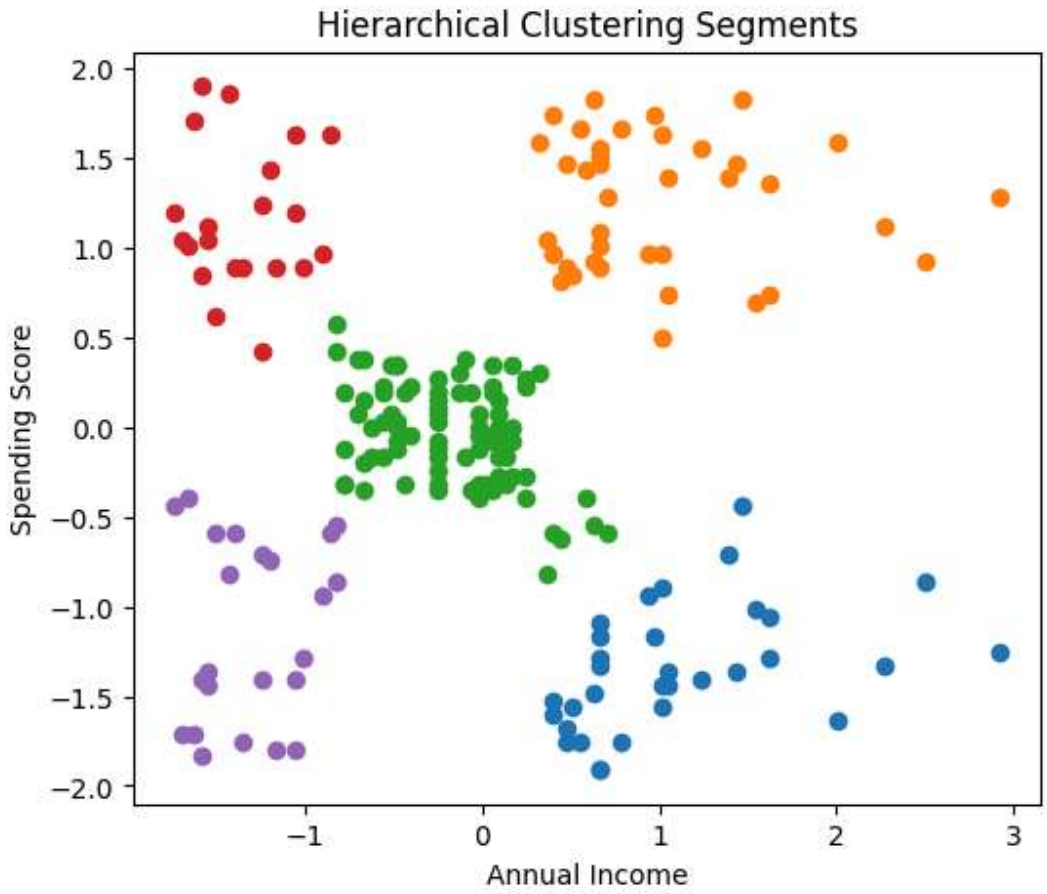
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.cluster import AgglomerativeClustering

# Load dataset
df = pd.read_csv('Mall_Customers.csv')

# Feature selection
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]

# Scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Elbow Method
wcss = []
for i in range(1,11):
   kmeans = KMeans(n_clusters=i, random_state=42)
```

```
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.plot(range(1,11), wcss)
plt.title("Elbow Method")
plt.xlabel("Clusters")
plt.ylabel("WCSS")
plt.show()

# K-Means
kmeans = KMeans(n_clusters=5, random_state=42)
y_kmeans = kmeans.fit_predict(X_scaled)

plt.scatter(X_scaled[:,0], X_scaled[:,1], c=y_kmeans)
plt.title("K-Means Clusters")
plt.show()

# Hierarchical Clustering
linked = linkage(X_scaled, method='ward')
dendrogram(linked)
plt.title("Dendrogram")
plt.show()

hc = AgglomerativeClustering(n_clusters=5)
y_hc = hc.fit_predict(X_scaled)

plt.scatter(X_scaled[:,0], X_scaled[:,1], c=y_hc)
plt.title("Hierarchical Clusters")
plt.show()
```

Output


Elbow Method
WCSS
Number of Clusters
400
350
300
250
200
150
100
50
2
4
6
8
10


Hierarchical Clustering Dendrogram
Distance
Customers
16
14
12
10
8
6
4
2
0



6. Performance Analysis

Since clustering is unsupervised, performance is evaluated using:

- Elbow Method (WCSS)
- Visual cluster separation

Observations:

- Elbow graph indicates optimal clusters around K = 5
- K-Means forms distinct and compact clusters
- Hierarchical clustering provides a tree-based structure
- Both methods successfully segment customers into groups

K-Means is efficient for large datasets, while hierarchical clustering provides better visualization.

7. Hyperparameter Tuning

K-Means Parameters:

- n_clusters (K)
- init (centroid initialization)
- max_iter

Tuning Method:

- Elbow Method used to determine optimal K

Impact:

- Correct K improves cluster separation
- Incorrect K leads to poor grouping

Hierarchical Clustering Parameters:

- n_clusters
- linkage method (ward, single, complete)

Tuning Method:

- Dendrogram analysis used to decide cluster count

Impact:

- Proper linkage improves cluster quality
- Wrong linkage leads to incorrect grouping

8. Conclusion

In this experiment, K-Means and Hierarchical clustering algorithms were applied to the customer segmentation dataset. The Elbow method helped determine the optimal number of clusters for K-Means, while the dendrogram was used in hierarchical clustering. Both techniques successfully grouped customers based on income and spending behavior. K-Means proved to be computationally efficient, whereas hierarchical clustering provided better visualization of cluster relationships. These clustering techniques are useful in real-world applications such as market segmentation and customer analysis.